Comparative Evaluation of 3D Hand Gesture Recognition Methods

MLSMM2154 - Artificial Intelligence Course

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Introduction

Context

- Rise of natural interfaces → traditional systems show their **limits**
- Gesture-based interfaces → **promising alternative**
- Depth sensors → 3D position vectors over discrete time

Gesture domains

- 1,000 sequences each (10 classes × 10 users × 10 reps)
- **Domain 1:** Arabic numerals 0–9 → **quasi-planar**
- **Domain 4:** CAD symbols → three-dimensional and **more complex**

Six methods

- **Baselines:** DTW, edit distance
- **Feature-based classifiers:** Random Forest, Decision Tree, Logistic Regression
- **Template matching:** $1 Recognizer adapted to 3D

Research Questions

RQ1: *What is the impact of preprocessing on classification performance?*

RQ2: *Which method achieves the best performance in user-dependent and user-independent settings?*

RQ3: *Are the performance differences between methods statistically significant?*

RQ4: *To what extent does the user-dependent protocol improve performance compared to user-independent?*

Experimental Setup

Cross-validation protocols

- **User-Independent:** recognise gestures of a user unknown at training time
- **User-Dependent:** recognise a new gesture from a known user
- Fold indices **shared** across all methods

Statistical tests (RQ3)

- Pairwise **Wilcoxon signed-rank** tests
- **Benjamini-Hochberg** correction → 15 pairs for 6 methods
- Significance threshold: $\alpha = 0.05$ after correction

Ablation study (RQ1)

- **(a)** No preprocessing
- **(b)** Standardisation alone
- **(c)** Standardisation + PCA denoising
- Best condition retained per method/protocol

Baseline Methods

Dynamic Time Warping

- **Measures similarity** by finding **optimal alignment** between two sequences
- Tolerates **differences in execution speed** between users
- Classification: 1-nearest-neighbour

Edit Distance

- 3D trajectories → **label sequences** via **k-means**
- Minimum number of insertions, deletions and substitutions
- Classification: 1-nearest-neighbour

$1 Recognize (3D)

Wobbrock and al. (2D) → canonical extension by Kratz and Rohs ($3)

1

Resample

- To exactly **150** equidistant points along its **arc length**
- To make all sequences **length-invariant**

2

Rotate

- Indicative **reference frame**
- Definition of a **reference direction** from the centroid to the first point
- **Rodrigues' rotation formula** to align gestures toward canonical orientation

3

Translate

- **Translation** of gesture
- To force its **centroid** at the **origin**

4

Scale

- **Uniform scaling**
- Gesture's largest **spatial dimension** equals 1
- **Scaling** on the **longest dimension** among the three axes

1-NN is then applied on the Golden Section Search (x, y, z): → $1 - d / (0.5 \cdot \sqrt{3} \cdot l^2)$

Feature-based classifiers

44 features (or 47 with EVR)

Random Forest

- **Number of trees :** 200
- **Splitted** on a random subset of $\sqrt{p}$ features
- **Bagging mechanism** decorrelates the trees and reduces variance.

Decision Tree

- Single tree
- **Objective :** measure the contribution of bagging

Logistic Regression

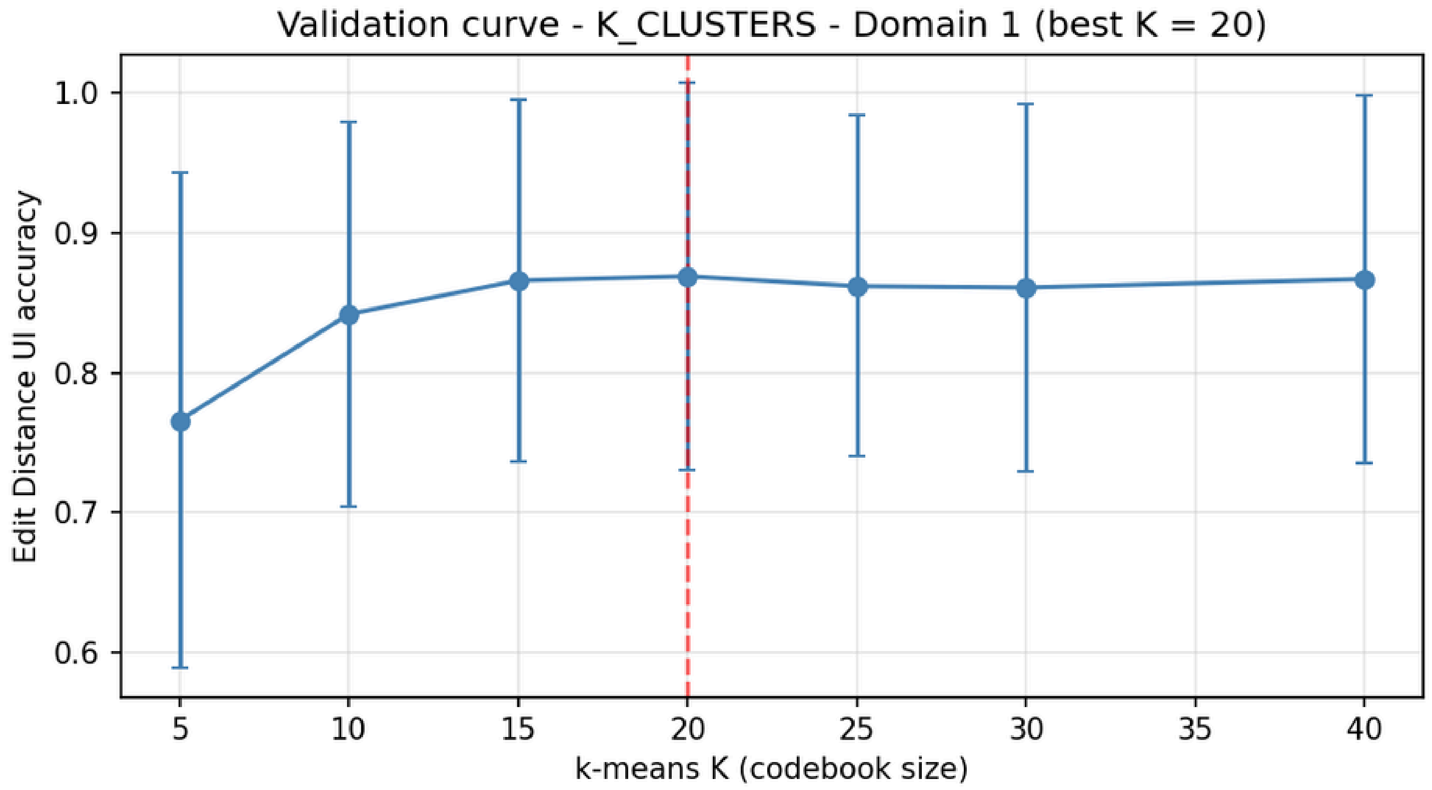
- **Linear baseline**
- **Comparison** with more complex models

Note: The LSTM was abandoned due to a structurally inefficient dataset, which led to underfitting.

Hyperparameters

Validation Curve (D1), K-means

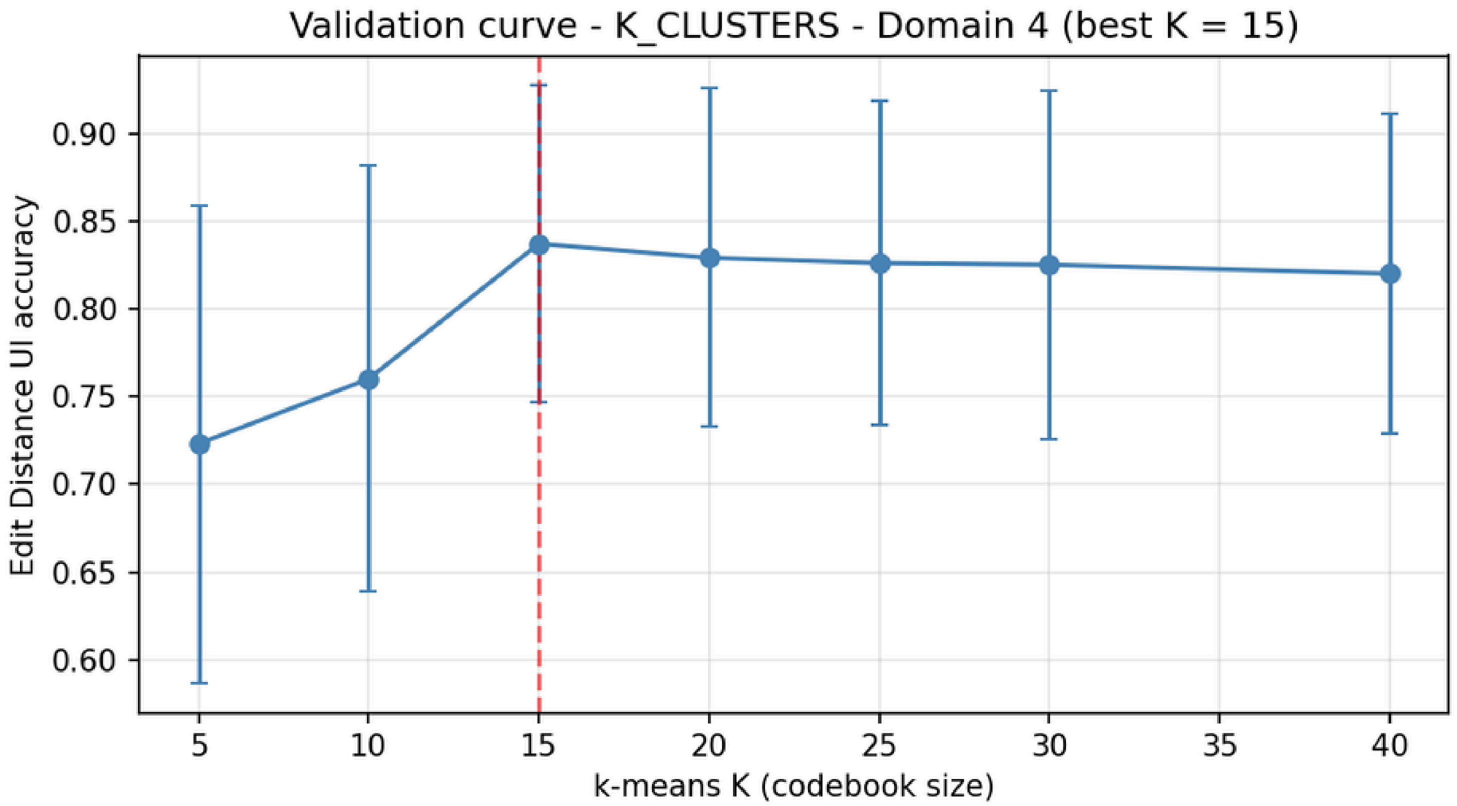
- Methods: Edit Distance only



Optimal : k = 20

Validation Curve (D4), K-means

- Methods: Edit Distance only

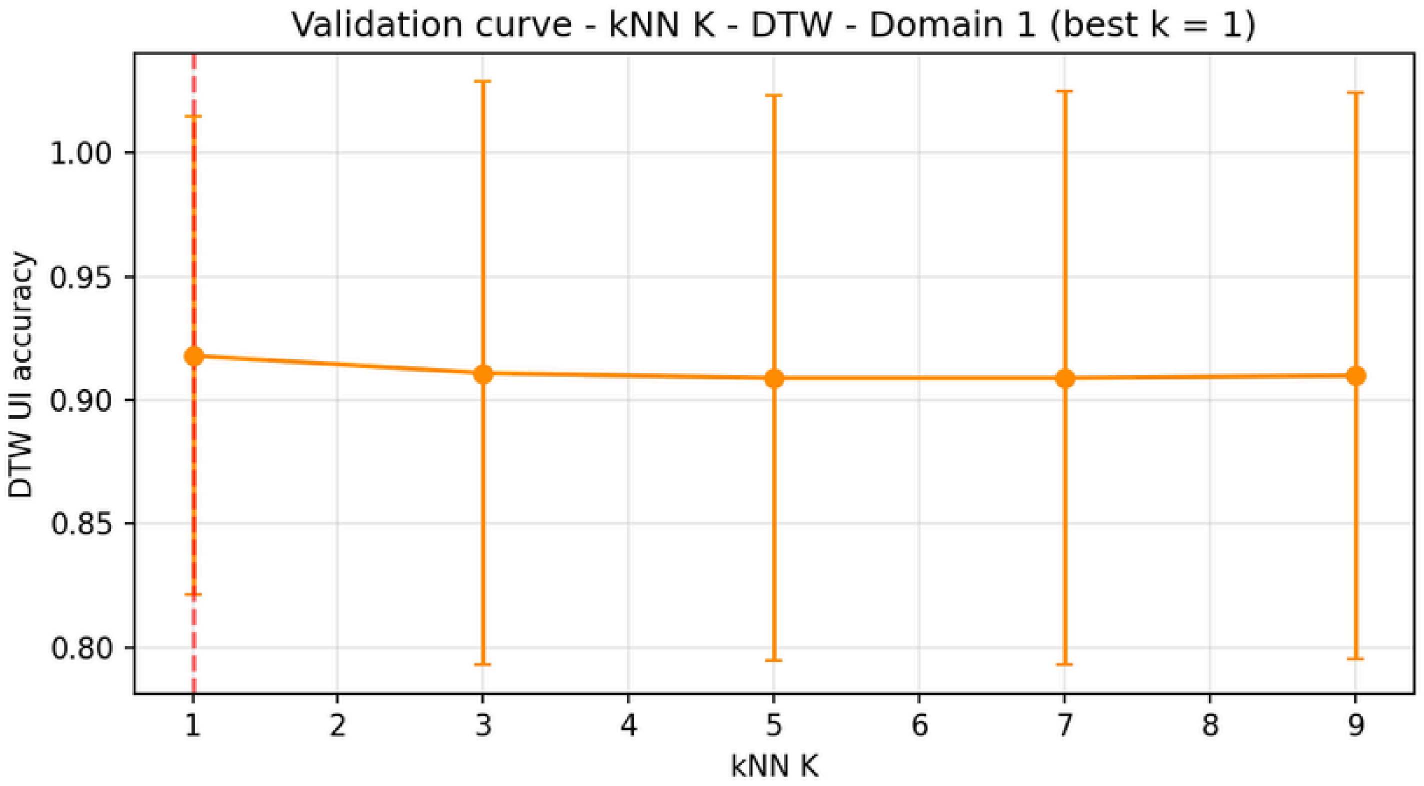


Optimal : k = 15

Hyperparameters

Validation Curve (D1), K-NN

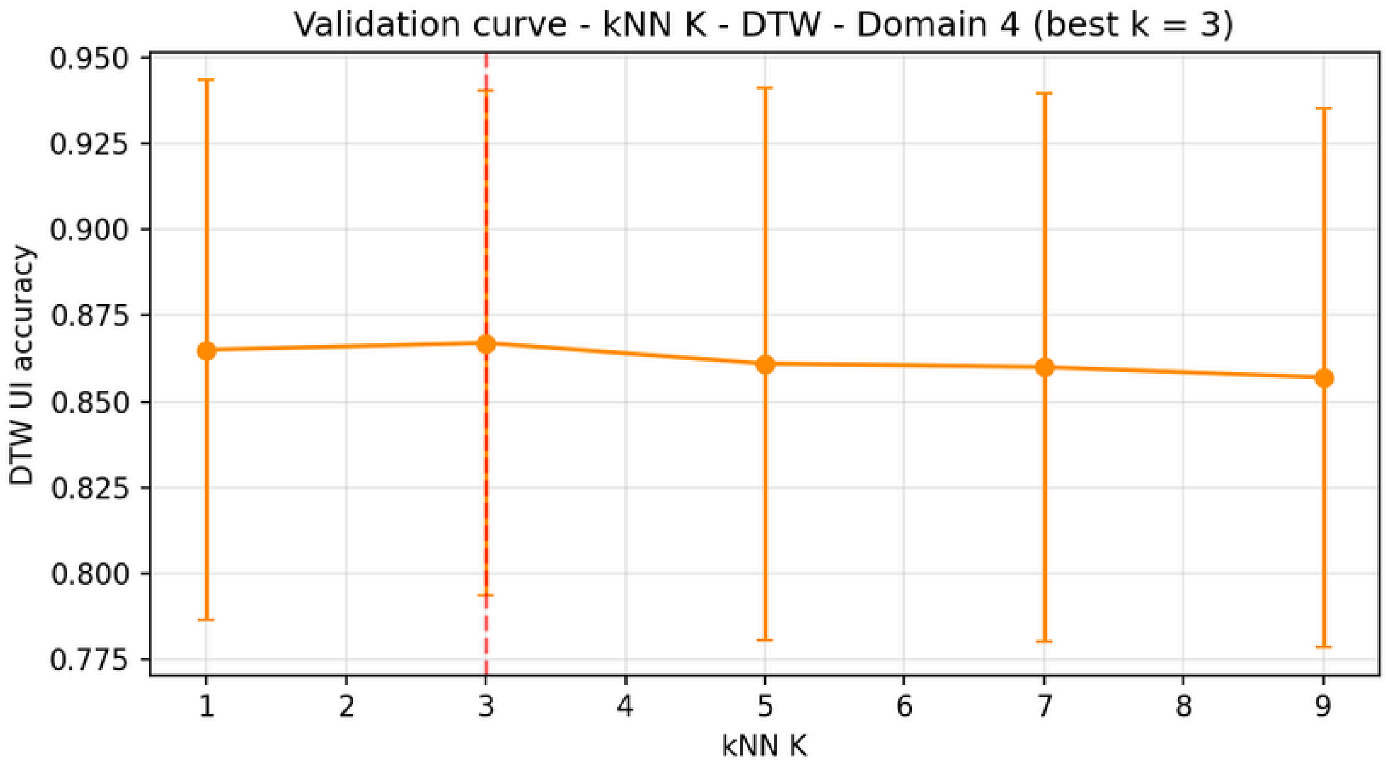
- **Methods: Edit Distance, DTW and $1**



Optimal : k = 1

Validation Curve (D4), K-NN

- **Methods: Edit Distance, DTW and $1**



Optimal : k = 3, but forced to 1 to stay consistent with standard practice in gesture recognition

Random Forest, Decision Tree and Logistic Regression use **GridSearchCV** with 5 inner folds, a nested cross-validation design. It helps to prevent test set **contamination** during tuning.

Results - Domain 1

Performances per preprocessing

Domain 1 - User Independent

Method	(a) Raw	(b) Std	(c) Std+PCA	Best
Edit Distance	0.552 ± 0.189	**0.869 ± 0.138**	0.828 ± 0.140	(b)
DTW	0.624 ± 0.167	**0.918 ± 0.097**	0.879 ± 0.108	(b)
DT	0.685 ± 0.068	**0.827 ± 0.081**	0.682 ± 0.111	(b)
RF	0.791 ± 0.095	**0.912 ± 0.055**	0.808 ± 0.059	(b)
LR	0.864 ± 0.171	**0.870 ± 0.093**	0.823 ± 0.112	(b)
$1	**0.992 ± 0.011**	0.885 ± 0.089	0.860 ± 0.080	(a)

Domain 1 - User Dependent

Method	(a) Raw	(b) Std	(c) Std+PCA	Best
Edit Distance	0.964 ± 0.031	**0.972 ± 0.022**	0.972 ± 0.019	(b)
DTW	0.978 ± 0.022	**0.988 ± 0.010**	0.967 ± 0.013	(b)
DT	0.825 ± 0.077	**0.875 ± 0.049**	0.766 ± 0.074	(b)
RF	0.949 ± 0.045	**0.956 ± 0.035**	0.888 ± 0.094	(b)
LR	0.955 ± 0.037	**0.956 ± 0.037**	0.917 ± 0.061	(b)
$1	**0.999 ± 0.003**	0.962 ± 0.013	0.935 ± 0.019	(a)

Domain 1 (UI) - Explanations

- **Standardisation** enhances performances for **all methods**, except **$1**.
- Adding **PCA** degrades performances across **all methods**, even despite **EVR3** = 5.8%

Domain 1 (UD) - Explanations

- **Hierarchy is preserved, $1 tops the board (99.9%)**
- Moderate UI-to-UD gains : geometrically well-separated digits, even across users
- Logistic Regression is not affected a lot by preprocessings due to its own internal StandardScaler

$1 Dominance: A pipeline precisely calibrated for these gestures, capturing **continuous geometric information** that other methods overlook.

Results - Domain 4

Performances per preprocessing

Domain 4 - User Independent

Method	(a) Raw	(b) Std	(c) Std+PCA	Best
Edit Distance	0.492 ± 0.149	**0.837 ± 0.090**	0.702 ± 0.085	(b)
DTW	0.486 ± 0.117	**0.865 ± 0.078**	0.752 ± 0.045	(b)
DT	0.692 ± 0.097	**0.813 ± 0.072**	0.573 ± 0.066	(b)
RF	0.827 ± 0.083	**0.864 ± 0.088**	0.718 ± 0.067	(b)
LR	**0.851 ± 0.105**	0.825 ± 0.112	0.708 ± 0.062	(a)
$1	0.821 ± 0.085	**0.831 ± 0.063**	0.705 ± 0.038	(b)

Domain 4 - User Dependent

Method	(a) Raw	(b) Std	(c) Std+PCA	Best
Edit Distance	0.945 ± 0.032	**0.984 ± 0.017**	0.949 ± 0.031	(b)
DTW	0.952 ± 0.029	**0.987 ± 0.020**	0.952 ± 0.028	(b)
DT	0.863 ± 0.041	**0.883 ± 0.026**	0.745 ± 0.072	(b)
RF	0.958 ± 0.025	**0.976 ± 0.021**	0.901 ± 0.048	(b)
LR	0.952 ± 0.030	**0.968 ± 0.018**	0.924 ± 0.040	(b)
$1	**0.973 ± 0.018**	0.956 ± 0.024	0.913 ± 0.029	(a)

Domain 4 (UI) - Explanations

- **Standardisation** remains essential with bigger improvements than with Domain 1
- **PCA** denoising is detrimental for **all methods:** PC3 retains **15.7%**
- **Logistic Regression** performs better on **raw data** thanks to its internal **StandardScaler**
- **$1** drops (0.992 to 0.831): its **2D-optimised pipeline** partly destroys the 3D structural cues that **distinguish volumetric symbols**
- **DTW**, operating on **raw 3D coordinates**, is more **robust**

Domain 4 (UD) - Explanations

- All **performances** converge above **0.88**
- The dominant **bottleneck** is **inter-user variability (UI)**, not geometric complexity

Note on $1 (UI): while **standardization** is the optimal preprocessing method, the result we kept (0.821) is based on **raw data**.

Results - Overview

Best performance by domain and by protocol

Dom.	Prot.	$1	DT	DTW	ED	LR	RF
1	UI	**0.992** ±0.011	0.827 ±0.081	0.918 ±0.097	0.869 ±0.138	0.870 ±0.093	0.912 ±0.055
	UD	**0.999** ±0.003	0.875 ±0.049	0.988 ±0.010	0.972 ±0.022	0.956 ±0.037	0.956 ±0.035
4	UI	0.821 ±0.085	0.813 ±0.072	**0.865** ±0.078	0.837 ±0.090	0.851 ±0.105	0.864 ±0.088
	UD	0.973 ±0.018	0.883 ±0.026	**0.987** ±0.020	0.984 ±0.017	0.968 ±0.018	0.976 ±0.021

Observations domain 1

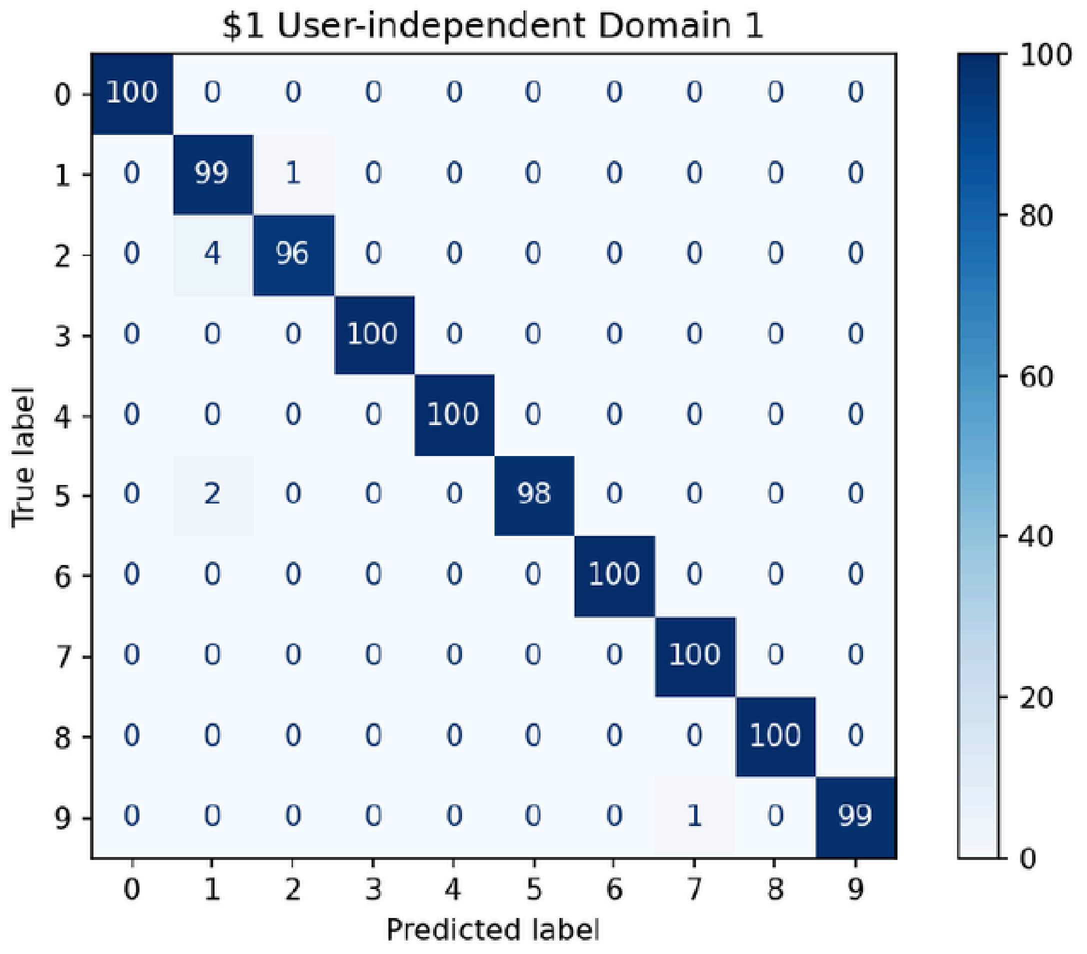
- D1 → clear winner : $1
- Closest competitors : RF

Observations domain 4

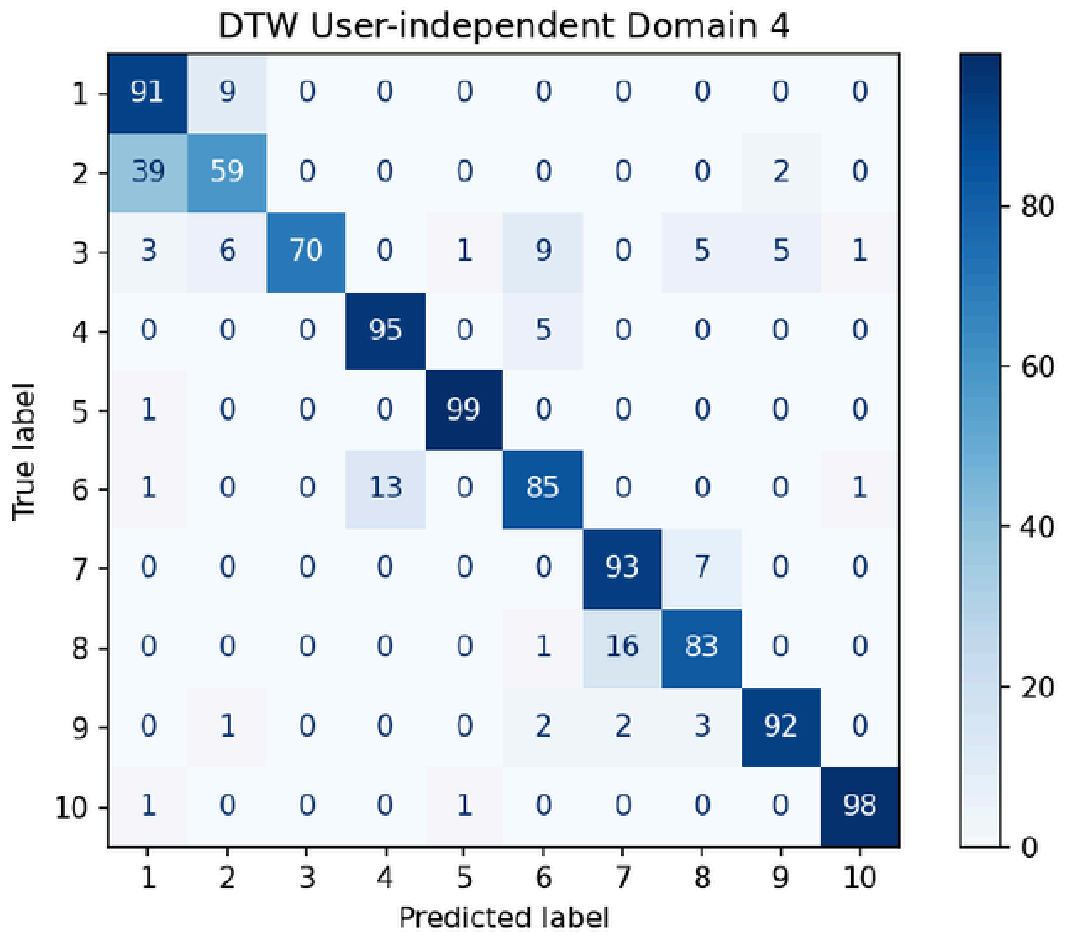
- D4 → no dominant method
- RQ4: UD gains up to +15.2 points on D4 → inter-user variability is the main bottleneck

Confusion Matrices

Confusion matrix $1



Confusion matrix DTW



Observations

- Quasi-diagonal → almost no errors
- 2 → 1 : 4 errors (vertical trajectory)
- 5 → 1 : 2 errors

Observations

- Cylinder → Cuboid : 39 errors (39%)
- Sphere → multiple classes : 30%
- Tetrahedron → Cyl. Pipe : 16 errors

Overfitting Analysis

Train-test gaps for classifiers (DT, RF, LR)

Domain	Protocol	DT (gap)	RF (gap)	LR (gap)
1	UI	**+0.128**	+0.080	+0.086
	UD	**+0.110**	+0.038	+0.033
4	UI	**+0.135**	**+0.124**	**+0.106**
	UD	**+0.106**	+0.023	+0.029

Observations

- $1, Edit Distance & DTW→ excluded
- DT → highest overfitting
- GridSearchCV → memorize
- RF → best balance (bagging)
- LR → most stable (L2 regularisation)

The key takeaway

- DT → overfits the most
- RF → best balance
- LR → stabe but limited

Statistical Significance

Paired Wilcoxon Signed-rank test (α = 0.05) : 15 pairs (6 methods) and a Benjamini-Hochberg correction

Domain 1

Method (A)	Edit Dist.	DTW	DT	RF	LR	$1
Edit Distance	–	0.0029	0.0943	0.2966	0.7411	0.0000
DTW	**0.0029**	–	**0.0013**	0.3475	0.0502	0.0018
DT	0.0943	0.0013	–	0.0000	0.0718	0.0000
RF	0.2966	0.3475	**0.0000**	–	**0.0264**	0.0000
LR	0.7411	0.0502	0.0718	0.0264	–	0.0000
$1	**0.0000**	**0.0018**	**0.0000**	**0.0000**	**0.0000**	–

- $1 **statistically better** than all methods
- DTW **statistically better** than Edit Distance, DT
- RF **statistically better** than DT, LR
- Other differences are **not significant**

Domain 4

Method (A)	Edit Dist.	DTW	DT	RF	LR	$1
Edit Distance	–	0.6313	0.6313	0.6313	0.7744	0.6603
DTW	0.6313	–	0.2498	0.9261	0.7178	0.2552
DT	0.6313	0.2498	–	0.0064	0.2552	0.8736
RF	0.6313	0.9261	**0.0064**	–	0.8583	0.4466
LR	0.7744	0.7178	0.2552	0.8583	–	0.4466
$1	0.6603	0.2552	0.8736	0.4466	0.4466	–

- 1 significant result : **RF statistically better** than DT
- All other comparisons are **not significant**
- DTW is the **best numerically** speaking
- **Numerically** better does not mean **statistically** better.

Conclusion & Limitations

Best Models

Domain 1 (UI & UD)
- $1 Recognizer is statistically better

Domain 4 (UI & UD)
- DTW (numerically) but not statistically significant

- Several methods are competitive

Limitations

- **Small dataset (N = 1,000)** → classical methods preferred

- **Asymmetric tuning** : GridSearchCV for RF/DT/LR, fixed params for DTW & Edit Distance

- **Preprocessing** is method-dependent : no universal pipeline

Resarch Questions

- **RQ1:** standardisation → +37.9 pts / PCA degrades

- **RQ4:** UD gains up to +15.2 pts on D4

Key Takeaways

- **Accuracy** alone is not enough

- **Also to consider**: overfitting, statistical significance, gesture domain nature

Thank you for listening

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Backup slides

PCA-denoising

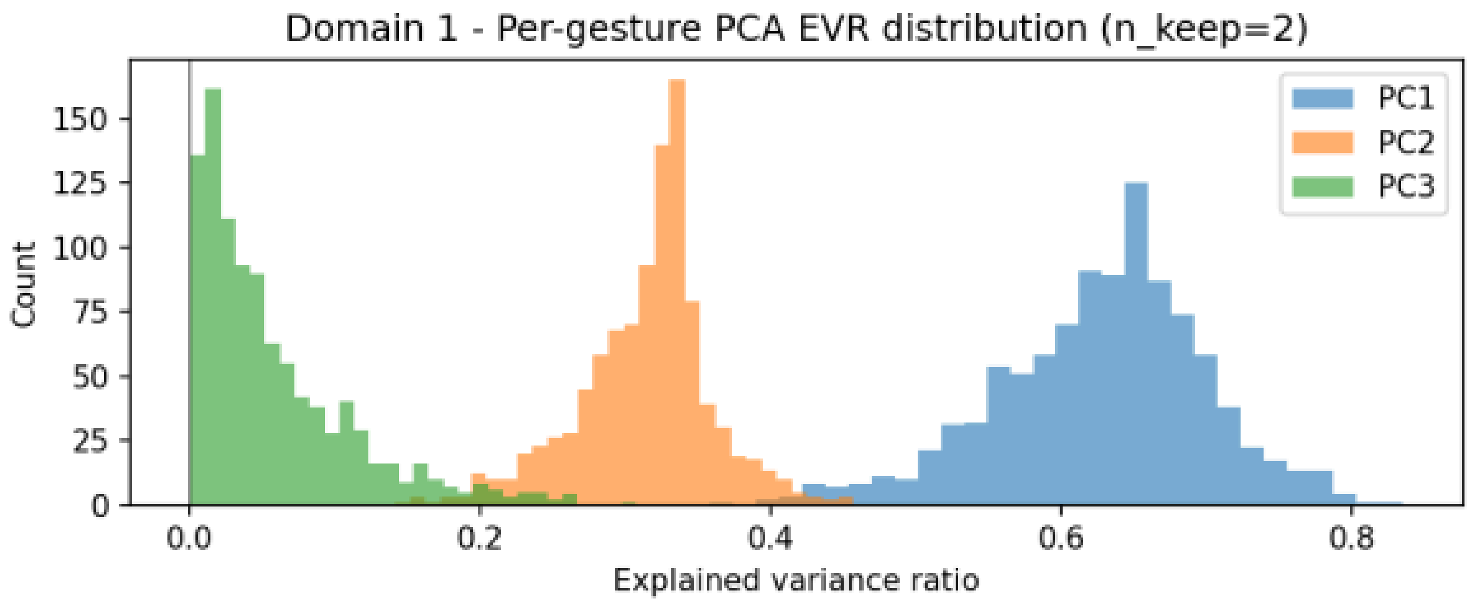

Figure 2: *Per-gesture PCA explained variance ratio (EVR) distribution for Domain 1.*

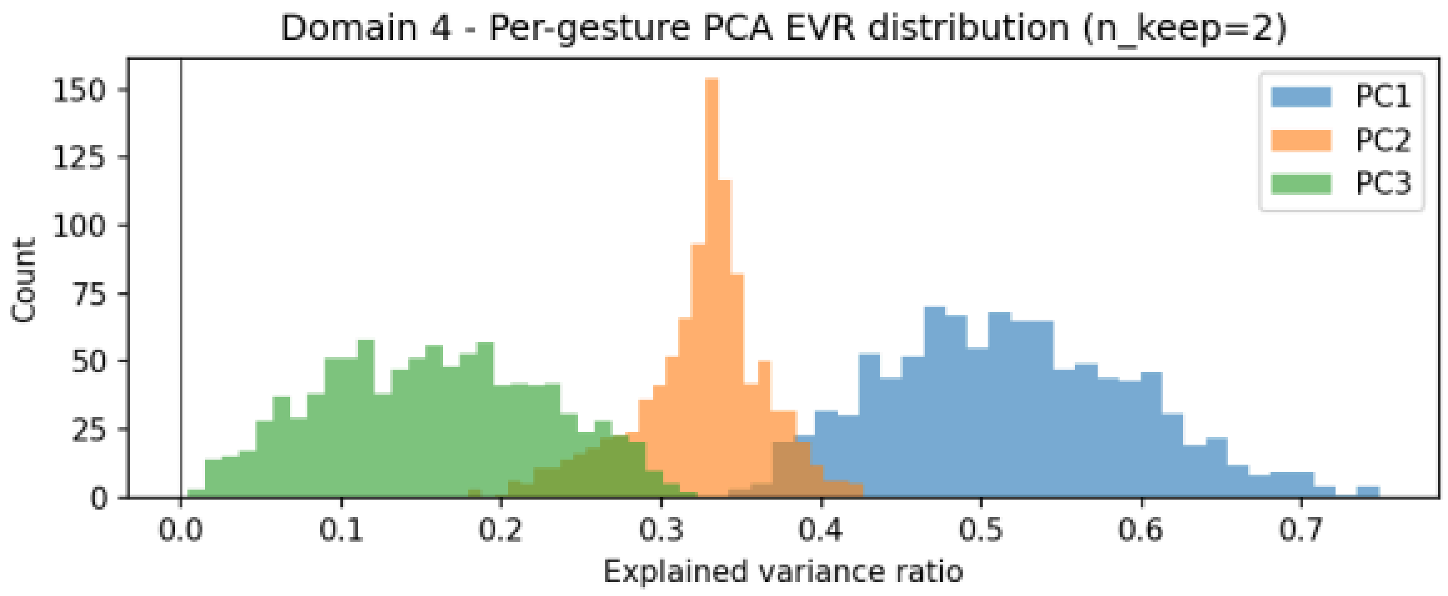

Figure 3: *Per-gesture PCA explained variance ratio (EVR) distribution for Domain 4.*

$$\mathrm{EVR}_j = \frac{\lambda_j}{\lambda_1 + \lambda_2 + \lambda_3}$$

Ablation study

Standardization by gesture, k-axis, instant t

$$\tilde{r}_k(t) = \frac{r_k(t) - \mu_k}{\sigma_k}$$

$$\mu_k = \frac{1}{T}\sum_{t=1}^{T} r_k(t), \qquad \sigma_k = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(r_k(t) - \mu_k)^2}$$

PCA-denoising

1. Empirical covariance matrix

$$\mathbf{\Sigma} = \frac{1}{T}\mathbf{X}^\top\mathbf{X} \in \mathbb{R}^{3\times3}$$

2. Projection 3D → 2D

$$\mathbf{Z} = \mathbf{X}\,\mathbf{V}_{1:2} \in \mathbb{R}^{T\times2}$$

3. Projection 2D → 3D

$$\tilde{\mathbf{X}} = \mathbf{Z}\,\mathbf{V}_{1:2}^\top \in \mathbb{R}^{T\times3}$$

$1 Recognizer

Pretreatment pipeline (4 stages)

Step 1

$$L_{\text{total}} = \sum_{t=1}^{T-1}\|\mathbf{p}_t - \mathbf{p}_{t-1}\|_2, \qquad \text{interval} = \frac{L_{\text{total}}}{N-1}$$

$$\mathbf{q} = \mathbf{p}_{i-1} + \frac{\text{interval} - D}{d}(\mathbf{p}_i - \mathbf{p}_{i-1})$$

Step 2

$$\theta = \arccos\left(\frac{\hat{\mathbf{p}}_1 \cdot \mathbf{c}}{\|\hat{\mathbf{p}}_1\|\,\|\mathbf{c}\|}\right), \qquad \mathbf{k} = \frac{\hat{\mathbf{p}}_1 \times \mathbf{c}}{\|\hat{\mathbf{p}}_1 \times \mathbf{c}\|}$$

$$\mathbf{v}_{\text{rot}} = \mathbf{v}\cos\theta + (\mathbf{k}\times\mathbf{v})\sin\theta + \mathbf{k}\,(\mathbf{k}\cdot\mathbf{v})(1-\cos\theta)$$

Step 3

$$\mathbf{r}'(t) = \mathbf{r}(t) - \mathbf{c}$$

Step 4

$$\text{max\_ext} = \max(x_{\max} - x_{\min},\ y_{\max} - y_{\min},\ z_{\max} - z_{\min}), \qquad \mathbf{r}''(t) = \mathbf{r}'(t)\cdot\frac{l}{\text{max\_ext}}$$

Classification

GSS

$$\varphi = \frac{\sqrt{5}-1}{2} \approx 0{,}618$$

$$c_1 = b - \varphi(b-a), \qquad c_2 = a + \varphi(b-a)$$

Distance (MSE)

$$d(S_{\text{cand}}, S_{\text{tmpl}}) = \frac{1}{N}\sum_{j=1}^{N}\|\mathbf{r}_{\text{cand}}(j) - \mathbf{r}_{\text{tmpl}}(j)\|_2^2$$

Confidence score

$$S = 1 - \frac{d}{0{,}5\cdot\sqrt{3}\cdot l^2}$$

Logistic regression

Multinomial softmax model

$$P(y=c \mid \mathbf{f}) = \frac{\exp(\mathbf{W}^{(c)} \cdot \mathbf{f} + b^{(c)})}{\sum_{c'=1}^{C} \exp(\mathbf{W}^{(c')} \cdot \mathbf{f} + b^{(c')})}$$

Cost function

$$\mathcal{L}(\mathbf{W}) = -\sum_{i=1}^{n} \log P(y_i \mid \mathbf{f}_i) + \frac{1}{2C} \|\mathbf{W}\|_F^2, \qquad \|\mathbf{W}\|_F^2 = \sum_{c,j} W_{cj}^2$$

Random Forest

$$\hat{y} = \arg\max_c \sum_{b=1}^{B} \mathbf{1}[h_b(\mathbf{f}) = c]$$

Decision Tree

$$\mathrm{Gini}(S) = 1 - \sum_{c=1}^{C} p_c^2$$

$$\mathrm{Entropy}(S) = -\sum_{c=1}^{C} p_c \log_2(p_c)$$

Wilcoxon signed-rank test for paired observations

$$d_i = a_i - b_i, \qquad W^+ = \sum_{i\,:\,d_i>0} \mathrm{rang}(|d_i|)$$

Features used by classifiers

Group	Nb	Description and key formula
Stats per axis x, y, z	21	Mean, std dev, min, max, range, skewness, kurtosis per axis (7 × 3)
Instantaneous speed	4	$v(t) = \|r(t) - r(t-1)\|_2$. Features: mean, std, max, min
Acceleration	2	$a(t) = \|v(t) - v(t-1)\|_2$. Features: mean, std
Arc length	1	$arc = \Sigma_t \|r(t) - r(t-1)\|_2$
Bounding box	4	Range per axis $(\Delta x, \Delta y, \Delta z)$ + diagonal $\|\Delta r\|_2$
3D curvature	3	$\varphi(t) = \arccos(\hat{v}(t) \cdot \hat{v}(t+1))$. Features: mean, std, sum
Speed per segment	3	Average speed in each temporal third
Displacement per segment	3	Average displacement per temporal third
Path per axis	3	$\Sigma_t \lvert r_k(t+1) - r_k(t)\rvert$ for $k \in \{x, y, z\}$
EVR PCA (optional)	3	$\lambda_j / \Sigma_k \lambda_k$ for $j \in \{1, 2, 3\}$

Instantaneous 3D curvature

$\hat{v}(t) = v(t) / \|v(t)\|_2$ $\quad$ $v(t) = r(t) - r(t - 1)$ $\quad$ $\varphi(t) = \arccos(\hat{v}(t) \cdot \hat{v}(t + 1))$